

MAT 310, Handout 27: Central Limit Theorem (7.2-7.4)

Theorem. Let $Y_1, Y_2, \dots, Y_n$ be independent and identically distributed (i.i.d.) $N(\mu, \sigma)$ RVs. Then,

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \sim$$

Note: we proved the general case of this using mgfs.

Example 1. Suppose the weight of apples shipped to a distributor is normally distributed with a mean of 8.4 oz. and a standard deviation of 1.0 oz. Find the probability that the mean of a sample of six apples will be at least 8 oz.

Question: What is the distribution of $\bar{Y}$ for i.i.d. Y_i that are not normal? We will use the following applet to investigate: https://digitalfirst.bfwpub.com/stats_applet/stats_applet_3_cltmean.html.

Central Limit Theorem (CLT): Let $Y_1, Y_2, \dots, Y_n$ be i.i.d. RVs with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2 < \infty$. Then, as $n \rightarrow \infty$,

$$\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \text{ converges in distribution to } N(0, 1).$$

In other words, $\bar{Y}$ is asymptotically normal with mean μ and variance σ^2/n .

Remark: Regardless of the distribution of the i.i.d. Y_i , the CLT implies that $\bar{Y}$ is approximately normal for large n . This is a powerful result! But how large is “large enough”?

Example 2. The weights of adult female elk are i.i.d. with a mean of 490 pounds and a standard deviation of 45 pounds.

- (a) Compute the probability that the average weight of 100 female elk is above 500 pounds.
- (b) Suppose that an elk relocation program is transporting 65 female elk to a new location. If the shipping freight capacity is 32,000 pounds, compute the probability that the total weight of the elk does not exceed the weight capacity.
- (c) Find the number of female elk that we would need to sample so that the absolute value of the difference between the sample mean of their weights and the population mean (490 pounds) is less than 5 pounds, with probability .90.

Theorem. Let $Y, Y_1, Y_2, \dots$ be random variables with moment-generating functions $m(t), m_1(t), m_2(t), \dots$, respectively. If $\lim_{n \rightarrow \infty} m_n(t) = m(t)$ for all real t , then the distribution function of Y_n converges to the distribution function of Y as $n \rightarrow \infty$.

Remark: This theorem is used to prove the CLT. If interested, the proof of the CLT is on pp. 377-378 of your textbook.